ADRENALIN: Multi-National Time-Series Dataset of Sub-Metered Building Energy

Authors:

Gustaf Bengtsson* (1), Rafael Gomez Garcia (1), Sofia Stensson (1), Matt Amos (2), Akram Hameed (2), Synne Krekling Lien (3), Igor Satori (3), Clayton Miller (4)

Affiliations:

(1) Built Environment, RISE Research Institutes of Sweden, Borås, Sweden

(2) CSIRO Energy Centre Newcastle, Commonwealth Scientific and Industrial Research Organisation, 10 Murray Dwyer Circuit, Mayfield West, NSW, 2304 Australia

(3) Sintef Community Pb, 124 Blindern, 0314 Oslo, Norway

(4) National University of Singapore, Singapore

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Building energy demand; Electricity; Time series data; Sub-meteringThermal loads

Data accessibility

Repository name: ADRENALIN - A building energy dataset [1]

Data identification number: 10.5281/zenodo.15012288

Direct URL to data: https://zenodo.org/records/15012288

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Abstract

The open dataset described in this paper contains several years of lowresolution building energy consumption data collected from 17 buildings located in different European countries, and Australia. Collectively, the 17 buildings comprise data from 371 energy meters, which sampled data at frequencies within the interval of minutes to one hour, resulting in approximately 15 billion data points. Originally created for the 'ADRENALIN 2024: Building Energy Load Disaggregation Challenge', the dataset contains time-series main and sub-meter data, corresponding to the total energy consumption (main meters for the whole-building), and to the weather-dependent heating and cooling loads (sub-meters), respectively. The dataset also contains contextual data of the buildings, together

with relevant weather data for the whole period. In addition to the variability in location, the dataset includes variability in the primary use of the buildings, including housing and non-residential.

Value of the Data

Beyond its primary application in energy disaggregation, the ADRENALIN dataset contians structured time-series data and building information that may support reuse in the following domains:

Algorithm Benchmarking: The dataset includes multi-year time-series data from 371 energy meters across 17 buildings, with varied sampling intervals and building types. This diversity enables comparative evaluation of load forecast models and load disaggregation methods.

Energy Flexibility and Demand Response: Sub-metered heating and cooling loads are recorded at granular intervals (5 minutes to 1 hour), allowing analysis of temporal energy patterns relevant to demand-side modelling and flexibility assessments.

Smart Control System Development: The dataset includes detailed thermal and electrical load profiles, which can be used to study control strategies for HVAC systems and other building energy systems.

Data structure and Accessibility: All data are provided in standardized formats (.csv and .txt), with consistent naming conventions and structure to support integration into analytic workflows.

Background

Buildings consume about 40% of global energy, making their efficient operation crucial for sustainability and reducing greenhouse gas emissions [2]. A major challenge in improving energy efficiency is the inability to break down total energy consumption into specific uses, such as heating, cooling, and lighting, from a single main meter. While sub-metering individual devices offers detailed insights, it is often too expensive and complex [3][4].

The ADRENALIN Load Disaggregation Challenge aimed to develop scalable, unsupervised machine learning algorithms to extract end-use energy patterns from aggregated data. This builds on previous work in energy disaggregation, such as non-intrusive load monitoring (NILM) techniques by Kelly and Knottenbelt [5] and Zeifman and Roth [6]. The challenge focused on methods that required minimal customization, ensuring they could be applied across various building types, time durations and climates.

A comprehensive dataset was created for this challenge, including several years of low-resolution energy consumption data, sampled with intervals ranging from 5 minutes to 1 hour, from buildings in Europe and Australia. It features time-series measurements from main and sub-meters, capturing total energy use and heating and cooling loads, along with metadata on building characteristics and climate variables. This dataset aims to advance energy disaggregation research and support broader scientific and practical efforts.

Data Description

Folder and file structure

The ADRENALIN dataset is organized by location, building, temporal resolution, meter, and metering source, following a structured key to decipher physical quantities and units via abbreviations. Each location comes with at least one set of building energy data and one set of weather data.

All building energy data and weather data are stored as comma-separated values (.csv) files, while building information and building meter topologies are saved as text files (.txt).

Combining the structure described above, each building energy data file is named based on the following format:

adrenalin.<temporal-resolution>.<location-id>.<building-id>.<meter-id>.<quantity-key>.csv

Example: adrenalin.15m.L01.B01.EM001.P.csv

Similarly, weather data files for each location follow this naming convention:

adrenalin.<temporal-resolution>.<location-id>.<weather station-id>.<quantity-key>.csv

Example: adrenalin.1h.L01.WS01.Rh.csv

Folders are organized by location, containing subfolders for weather data and formatted time-series data sorted by various sampling resolutions (e.g., 1 hour, 30 minutes, 15 minutes, or 5 minutes). Note that not all buildings provide resolutions higher than 1 hour. Additionally, a folder containing raw building energy data accompanies reformatted data, preserving the data as received from the building management system. Figure 1, below, illustrates an example of the hierarchical directory structure L03.zip, with the remainder of the files following the same structure.

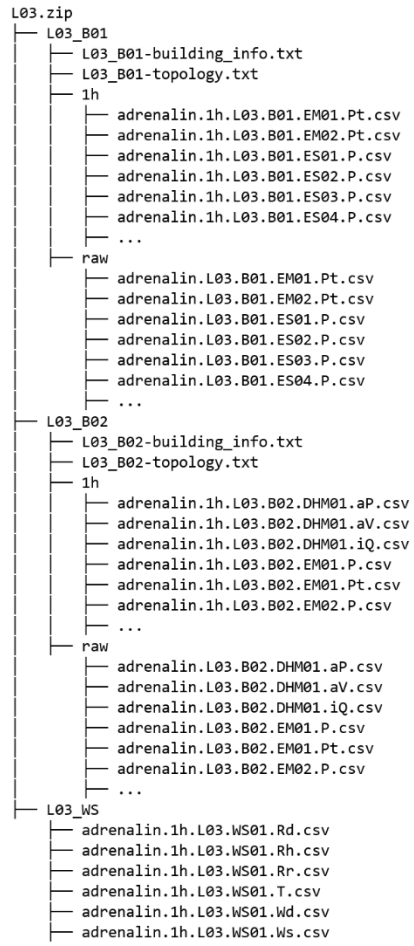


Figure 1 Example of hierarchical directory structure of L03.zip, illustrating folder and file structure and their hierarchical relationships.

Meter IDs are designed to provide detailed information about the energy carrier a meter monitors (e.g., electricity, district heating or cooling), whether it is a main meter or a sub-meter, and its position in the building energy management hierarchy, which corresponds to the building topology.

The naming convention begins with a prefix that defines the meter's role in the building's energy system (e.g., electricity as "E," district heating as "DH"). This is followed by a main meter (M) or sub-meter (S) identifier, and a suffix serial number to indicate the specific meter's position. For example, in "adrenalin.15m.L01.B01.EM001.Rh.csv," the "EM001" refers to the first Electricity Main meter located at the top of the hierarchy. A summary of the abbreviations and meter types is as follows:

Table I. Abbreviations of occurring energy carriers and resulting combinations of meter-ids used as identifiers in dataset.

Carrier	***Abbreviation***	***Meter-IDs***

Electricity	E	EM, ES
District Heating	DH	DHM, DHS
District Cooling	DC	DCM, DCS
Heating	H	HM, HS,
Cooling	C	CM, CS
Gas	G	GM, GS
Free Cooling	FC	FCM, FCS

This system ensures a consistent and logical framework for identifying and managing energy meters within the dataset.

File details

All building and weather data files are structured as timeseries data, meaning they consist of measurements recorded at specific moments in time, allowing for trend analysis and monitoring over a given period. Each file includes two columns:

1. Timestamp - Each entry represents a specific moment in time. The timestamps are formatted using ISO 8601 standards (e.g., 2020-10-16T00:00:00Z), where the Z indicates that the time is given in Coordinated Universal Time (UTC).
2. Value - Each timestamp corresponds to a numerical value. These values represent measurements based on the quantity-key indicated in the filenames.

The column labelled Value provides details about the physical quantity being recorded, as well as the unit of measurement. For instance:

- P represent power consumption, recorded by a meter.
- eP represent exported power, where the initial e denotes export or generation.

These examples illustrate that the dataset captures both imported and exported energy flows. It is important to note that these values are recorded independently and represent gross quantities of a total drawn by the building, and exports sent back to the grid or generation on-site. Users interested in net power flow can compute it by subtracting export from total loads where applicable.

A complete list of physical quantities based on abbreviations used is found in the file “ADRENAL – data keys”.

The total number of rows in each file depends on the duration over which the meter collected data and the resolution of the recordings. For example, higher-resolution files (e.g., recorded every 15 minutes) will have more rows compared to lower-resolution files (e.g., hourly recordings).

Table II. Sample of three examples of datafiles containing time-series data from the dataset. Each datafile is Identified by file name in the column header followed by the first 10 row plus column names in the .csv file.

File: *adrenalin.L06.B01.EM01.P*		File: *adrenalin.L06.B01.EM01.eP*		File: *adrenalin.30m.L06.B01.EM01.P*	
timestamp [UTC]	value P [Total active power (kW)]	timestamp [UTC]	value eP [Active power export (kW)]	timestamp [UTC]	value P [Total active power (kW)]
2020-10-02T09:48:59Z	0	2020-10-02T09:48:59Z	0	2020-10-02T09:30:00Z	0
2020-10-02T09:49:00Z	0	2020-10-18T11:14:09Z	0.1	2020-10-12T12:00:00Z	0.650000036
2020-10-12T12:00:09Z	0.6	2020-10-18T11:18:02Z	0	2020-10-12T12:30:00Z	0.606666684
2020-10-12T12:01:09Z	0.7	2020-10-18T11:19:29Z	0.1	2020-10-12T13:00:00Z	0.62499994
2020-10-12T12:02:18Z	0.6	2020-10-18T11:20:40Z	0	2020-10-12T13:30:00Z	0.620000005
2020-10-12T12:03:42Z	0.7	2020-10-18T11:21:43Z	0.1	2020-10-12T14:00:00Z	0.646153808
2020-10-12T12:05:11Z	0.6	2020-10-18T11:22:45Z	0	2020-10-12T14:30:00Z	0.683333397
2020-10-12T12:06:15Z	0.7	2020-10-18T11:24:29Z	0.1	2020-10-12T15:00:00Z	0.653333306
2020-10-12T12:07:31Z	0.6	2020-10-18T11:25:32Z	0	2020-10-12T15:30:00Z	0.688888848
2020-10-12T12:08:51Z	0.7	2020-10-18T11:29:41Z	0.1	2020-10-12T16:30:00Z	0.799999952

Metadata

Each building is accompanied by metadata in a building information text file, including:

- **Building characteristics:** e.g., primary building use, climate zone, HVAC systems.
- **Energy system attributes:** e.g., meter types, number of submeters.

The dataset includes a meter topology schema mapping relationships between main meters and submeters for each building.

For instance, the topology captures hierarchical relationships, such as one main meter distributing power to submeters for lighting, HVAC, and plug loads. The following Figure 2, illustrates the schematic of meter topology for one building.

```
GROCERY STORE (L07_B01)

EM01 (Electricity main meter)
|
|- ES01 (Fresh produce)
|- ES02 (Lights, sockets, POS)
|- ES04 (Energy group meter)
|    |- ES05 (Chiller)
|    |- ES06 (Rooms and disks)
|    |- ES07 (Dry cooler)
|- ES03 (Air handling unit) [TD]

NOTE:
[TD] = temperature dependent load.
```

Figure 2. Example of a building topology, i.e. building L07_B01.

Experimental Design, Materials and Methods

Data acquisition

Energy consumption data were collected as part of the ADRENALIN project, involving a consortium of data providers managing buildings across Europe and Australia. The data collection process was initially guided by inclusion criteria for the competition:

1. Buildings were required to have separate energy meters for heating and cooling systems, with sampling intervals of one hour or less.
2. Buildings were expected to rely on electrical energy for thermal loads (heating/cooling), excluding those primarily using district heating or cooling systems.

However, despite these exclusion criteria, data collected from the available affected buildings was nevertheless retained. This decision was made to ensure the availability of these data for the research community after the competition. By including these additional data, the dataset offers broader opportunities for diverse research applications.

Building metadata, including attributes such as data provider, building type, floor area (gross heated/cooled area), number of stories, HVAC system characteristics, and climate zone classification, were obtained to ensure comprehensive context.

Meteorological weather data time-series for the geographical locations of the buildings were obtained from publicly available sources, specifically the CERRA [8], ERA5-Land hourly data [9], and ERA5 hourly data on single levels [10] datasets. To comply with data protection regulations, particularly the General Data Protection Regulation (GDPR), precise geographic coordinates and weather station identifiers were excluded from the dataset. Instead, climate zone classifications and reformatted anonymized weather time series (.csv) were provided to preserve privacy while maintaining the dataset's reusability.

The collection process adhered to robust data security protocols, utilizing encrypted transfers and designated landing zones for data storage.

Project partners

- ReMoni (Denmark), Synavision (Germany) and Kiona (Norway):
  These partners contributed energy data collected by their systems installed to monitor energy use in buildings across Denmark, Germany, and Norway. The resolution and number of meters varied based on customer demands for energy consumption surveillance and management. Data was handed over in the summer of 2022 and covers various periods from 2017 to 2023. Raw data was extracted from each partners database and anonymized, before being processed and harmonized.
- CSIRO (Australia):
  Contributed data from three office buildings situated on the East Coast of Australia, with each building representing a different latitude. Data from three buildings includes energy generation and consumption data recorded between January 2021 and January 2023. The data was collected by Building Management Systems (BMS) within each building and ingested into CSIRO's Data Clearing House (DCH) data platform. The Brick schema was utilised to describe metadata about the types of data and equipment within the buildings onboarded. Raw data was extracted from the Data Clearing House in August 2023 and anonymized before being subjected to the ADRENALIN dataset pre-process, harmonization and validation procedure.

- SINTEF (Norway):
  Contributed energy consumption data from five non-residential buildings owned by Drammen Municipality in Norway. The energy consumption of all public buildings in Drammen is monitored through an energy surveillance system. The raw data for these five buildings were downloaded in February/March 2022 and processed alongside the dataset described in https://zenodo.org/records/11060089. After downloading the data, each meter was manually inspected and re-labelled. The dataset underwent cleaning to remove false zeros, negative values, outliers beyond specified thresholds, and days with missing data [7]. Data from SINTEF was processed and harmonized to fit in the ADRENALIN dataset.

A summary of the data providers, building locations, climates, and metering configurations is presented in the following table.

Table III. Overview of building from which meter data was collected. 1 Köppen climate classification system; 2 Total area in square meters (total heated area within parentheses); 3 Ventilation system (VS), Heating & cooling system (H&CS), Tap water (TW); 4 Electricity Main meter (EM), Electricity Submeter (ES), Virtual Main meter (VM), Virtual Submeter (VS), District Heating Main meter (DHM), Heat Main meter (HM), Heat Submeter (HS); 5 The building has more meters whose recordings are not included in the data sets; 6 Largest number of stories (the building has other sections (e.g., wings) with fewer number of storeys/floors).

Country	***Building ID***	***Characteristics***	
Australia	L01_B01	Data Provider: CSIRO Climate Zone[1]: Cfb Year Opened: 2017	Nr Storeys: 5 Area[2]: 15000 (15000) Primary Use: Commercial (Office)
		Energy[3]: VS: used for space heating & cooling; TW: natural gas; Solar PV Nr of meters [4, 5]: 2 EM, 45 ES Aggregated file size: 9.29 GB Temporal coverage: 2021 January - 2022 December Resolutions: 1h, 30m, 15m, 5m Units: kW, kWh, A, V, kVAR, kVA, Hz	
	L02_B01	Data Provider: CSIRO Climate Zone[1]: Cfb Year Opened: 2005	Nr Storeys: 3 Area[2]: 2600 (1500) Primary Use: Commercial (Office)

		Energy[3]: VS: used for space heating & cooling; TW: natural gas; Solar PV Nr of meters [4, 5]: 1 EM, 6 ES Aggregated file size: 3.79 GB Temporal coverage: 2021 January - 2022 December Resolutions: 1h, 30m, 15m, 5m Units: kW, kWh, A, V, kVAR, kVARh, kVA, Hz	
	L09_B01	Data Provider: CSIRO Climate Zone[1]: Cfa Year Opened: 2003	Nr Storeys: 4 [6] Area[2]: 2600 (1500) Primary Use: Commercial (Office)
		Energy[3]: VS: used for space heating & cooling; TW: natural gas; Solar PV Nr of meters [4, 5]: 1 EM, 37 ES Aggregated file size: 12. 9 GB Temporal coverage: 2021 January - 2022 December Resolutions: 1h, 30m, 15m, 5m Units: kW, kWh, A, V, kVAR, kVA, Hz	
Denmark	L03_B01	Data Provider: ReMoni Climate Zone[1]: Cfb Year Opened: 2013	Nr Storeys: 4 Area[2]: 1013 (1013) Primary Use: Institutional (Edu.)
		Energy[3]: VS: used for space heating; H&CS: gas boiler; TW: gas boiler, district heating Nr of meters [4]: 1 VM, 1 EM, 4 ES, 5 HS Aggregated file size: 9.38 MB Temporal coverage: 2022 January - 2022 December Resolutions: 1h Units: kW, kWh, m^3, deg C	
	L03_B02	Data Provider: ReMoni Climate Zone[1]: Cfb	Nr Storeys: missing Area[2]: missing

		Year Opened: missing	Primary Use: missing
		Energy[3]: missing Nr of meters [4]: 1 EM, 1 ES, 1 DHM Aggregated file size: 25.2 MB Temporal coverage: 2020 October - 2022 December Resolutions: 1h Units: kW, kWh, m^3, deg C	
Germany	L05_B01	Data Provider: Synavision Climate Zone[1]: Cfb Year Opened: 2010	Nr Storeys: 3 Area[2]: 6700 (6700) Primary Use: Institutional (Governmental)
		Energy[3]: VS: used for space heating & cooling; H&CS: electric heating, cooling machine, geothermal storage, floor heating/cooling Nr of meters [4]: 1 EM, 39 ES, 1 HM, 5 HS Aggregated file size: 270 MB Temporal coverage: 2007 May - 2008 August Resolutions: 1h, 30m, 15m Units: kW, kWh	
	L05_B02	Data Provider: Synavision Climate Zone[1]: Cfb Year Opened: 2010	Nr Storeys: 3 Area[2]: 6689 (6689) Primary Use: Institutional (Education)
		Energy[3]: H&CS: heat pump, floor heating/cooling, backup gas boiler; TW: boiler; Solar PV Nr of meters [4]: 1 EM, 39 ES Aggregated file size: 174 MB Temporal coverage: 2018 July - 2023 December Resolutions: 1h, 30m, 15m	

		Units: kW, kWh, m^3	
	L08_B01	Data Provider: Synavision Climate Zone[1]: Cfb Year Opened: 2015	Nr Storeys: 4 Area[2]: 11500 (11500) Primary Use: Institutional (Education)
		Energy[3]: H&CS: heat pump, geothermal storage, agrothermics; Solar PV Nr of meters [4]: 2 EM, 9 ES, 4 HS Aggregated file size: 72.5 MB Temporal coverage: 2016 February - 2017 June Resolutions: 1h, 30m, 15m, 5m Units: kW, kWh	
Norway	L06_B01	Data Provider: Kiona Climate Zone[1]: Cfb Year Opened: 2020	Nr Storeys: 2 Area[2]: 2200 (2200) Primary Use: Commercial (Retail)
		Energy[3]: VS: used for space heating & cooling; H&CS: heat pump; TW: heat pump, electric accumulation tank; Solar PV Nr of meters [4]: 2 EM, 2ES, 1 HM, 2HS Aggregated file size: 1.03 GB Temporal coverage: 2020 October - 2022 December Resolutions: 1h, 30m, 15m, 5m Units: kW, kWh, A, V, kVAR, kVARh, kVA, m^3, deg C	
	L07_B01	Data Provider: Kiona Climate Zone[1]: Cfb Year Opened: 1999	Nr Storeys: 1 Area[2]: 450 (450) Primary Use: Commercial (Grocery Store)
		Energy[3]: VS: used for space heating & cooling Nr of meters [4]: 1 EM, 7 ES	

		Aggregated file size: 1.74 GB Temporal coverage: 2021 November - 2023 January Resolutions: 1h, 30m, 15m, 5m Units: kW, kWh, A, V, kVAR, kVARh, kVA, Hz	
	L10_B01	Data Provider: Kiona Climate Zone[1]: Cfb Year Opened: 2013	Nr Storeys: 6 Area[2]: 1800 (1800) Primary Use: Commercial (Office)
		Energy[3]: H&CS: district heating, district cooling, free cooling Nr of meters [4]: 2 EM, 23 ES Aggregated file size: 6.01 GB Temporal coverage: 2019 March - 2023 January Resolutions: 1h, 30m, 15m, 5m Units: kW, kWh, A, kVAR, kVARh, kVA, Hz, m^3, deg C	
	L11_B01	Data Provider: Kiona Climate Zone[1]: Cfb Year Opened: 2016	Nr Storeys: 10 Area[2]: 3684 (3684) Primary Use: Residential (Multi-family)
		Energy[3]: VS: used for space cooling; H&CS: district heating, district cooling, electric heating; TW: electric heating Nr of meters [4]: 2 EM, 15 ES Aggregated file size: 8.29 GB Temporal coverage: 2019 December - 2023 January Resolutions: 1h, 30m, 15m, 5m Units: kW, kWh, A, kVARC, kVARL, kvVARhC, kVARhL, ...	
	L11_B02	Data Provider: Kiona Climate Zone[1]: Cfb Year Opened: 2019	Nr Storeys: 1 Area[2]: 3000 (2469) Primary Use: Commercial (Office)

<table>
<tr><td></td><td></td><td colspan="2">Energy[3]: VS: used for space heating; TW: electric heating
Nr of meters [4]: 2 EM, 15 ES
Aggregated file size: 2.22 GB
Temporal coverage: 2019 December - 2022 December
Resolutions: 1h, 30m, 15m, 5m
Units: kW, kWh, V, A, kVAR, kVARh, Hz, ...</td></tr>
<tr><td></td><td>L14_B01</td><td>Data Provider: SINTEF
Climate Zone[1]: Cfb
Year Opened: 1914
(renovated 2001)</td><td>Nr Storeys: 5
Area[2]: 6084 (6084)
Primary Use: Institutional (Education)</td></tr>
<tr><td></td><td></td><td colspan="2">Energy[3]: VS: used for space heating & cooling; H&CS: electric boiler, heat pump; TW: heat pump & electric boiler
Nr of meters [4]: 1 EM, 1 ES, 1 HM
Aggregated file size: 9.22 MB
Temporal coverage: 2017 December - 2022 March
Resolutions: 1h
Units: kWh</td></tr>
<tr><td></td><td>L14_B02</td><td>Data Provider: SINTEF
Climate Zone[1]: Cfb
Year Opened: missing</td><td>Nr Storeys: missing
Area[2]: 2700 (2700)
Primary Use: Service (Nursing home)</td></tr>
<tr><td></td><td></td><td colspan="2">Energy[3]: VS: used for heating & cooling; H&CS: electric boiler, heat pump; TW: heat pump & electric boiler
Nr of meters [4]: 1 EM, 3 ES
Aggregated file size: 11.4 MB
Temporal coverage: 2017 December - 2022 February
Resolutions: 1h
Units: kWh</td></tr>
</table>

	L14_B03	Data Provider: SINTEF Climate Zone[1]: Cfb Year Opened: 2010	Nr Storeys: 3 Area[2]: 6574 (6574) Primary Use: Institutional (Education)
		Energy[3]: VS: used for heating & cooling; TW: district heating, electric heating accumulation tank Nr of meters [4]: 2 EM, 6 ES, 1 HM, 2 HS Aggregated file size: 20.3 MB Temporal coverage: 2017 December - 2022 March Resolutions: 1h Units: kWh	
	L14_B04	Data Provider: SINTEF Climate Zone[1]: Dfb Year Opened: 1871	Nr Storeys: 3 Area[2]: 11502 (11502) Primary Use: Commercial (Office)
		Energy[3]: VS: used for space heating; H&CS: district heating, heat pump, TW: district heating, electric heating accumulation tank Nr of meters [4]: 2 EM, 2 ES, 1 HM, 2 HS Aggregated file size: 12.3 MB Temporal coverage: 2017 December - 2022 March Resolutions: 1h Units: kWh	
	L14_B05	Data Provider: SINTEF Climate Zone[1]: Dfb Year Opened: 1990	Nr Storeys: 2 Area[2]: 988 (988) Primary Use: Service (Kindergarten)
		Energy[3]: VS: used for space heating; H&CS: district heating; TW: electric heating accumulation tank Nr of meters [4]: 1 EM, 1 HM, 1 HS Aggregated file size: 7.47 MB	

		Temporal coverage: 2017 December - 2022 March Resolutions: 1h Units: kWh

Data Pre-Processing and Harmonization

To create a consistent and uniform dataset, a rigorous harmonization process was applied to the collected raw data:

- **Reformatting and Standardization:** Temporal inconsistencies were addressed by synchronizing all datasets to a uniform timestamp format, including adjustments for time zones and daylight-saving time. Variables across datasets were standardized in terms of naming conventions, units, and structure.
- **Resampling:** Event-triggered recordings of values were converted to a frequency-based format to ensure versions of timeseries with uniformly aligned data points over time.
- **Missing Values:** Missing or corrupt data were retained as NaNs (Not a Number, a placeholder used in data science to represent missing or undefined values) to preserve the integrity of the original measurements, avoiding imputation techniques.
- **Variable Integration:** A common set of variables representing key energy and thermal metrics was defined. Missing variables in the raw data were filled with NaN values to maintain uniformity.
- **Virtual Meter Creation:** For buildings with separate heating and cooling meters, virtual meters were generated to aggregate thermal loads. Aggregation was performed at the lowest existing temporal resolution to ensure data integrity. Virtual meters were computed using summation algorithms, with safeguards against anomalies through pre-aggregation filtering.
- **File Structuring:** Filenames were formatted to include key identifiers such as building location, meter type, and recorded variables, ensuring traceability and facilitating dataset navigation.

Metadata and Meter Topology

During the Data acquisition phase, metadata for each building was collected using standardized templates and structured interviews with project partners. These templates were designed to capture essential building characteristics and energy system configurations, and the resulting structure reflects the organization of the collected

information. This approach ensured consistency across diverse sources but does not represent a formal metadata schema or ontology.

For the CSIRO buildings, metadata was extracted from internal models that utilize the Brick schema; however, only selected attributes were included in the dataset, as the primary focus was on time-series energy data. For other buildings, the available metadata was not sufficient to support the use of standardized ontologies.

This collaborative process, involving iterative checks and validation, ensured comprehensive and reliable metadata. The resulting topology schema maps relationships between main meters and submeters, supporting energy flow analysis and load disaggregation algorithms.

Data quality assessment

A data quality assessment was employed to unveil irregularities and detect outliers in timeseries available in the dataset. Metrics were aggregated at both the meter level and the building or weather station level to provide an overview of the results in a more accessible manner.

Temporal consistency checks were conducted to identify irregularities in timestamps, including missing values (MISS), non-monotonic sequences, and inconsistent sampling intervals (TSI). The occurrence of intervals between timesteps deviating from a timeseries' expected temporal resolution were counted and stored as inconsistent sampling intervals. Thresholds of 4 times timeseries resolution and gaps longer than 1 day were used. Special consideration was given to event-based meters, which log data only when values change. These were identified by analysing the variability and maximum gaps between timestamps. Meters with high interval variance and gaps over one day were excluded from interval-based scoring to avoid penalizing intentional event-driven behaviour.

Outlier detection was performed using two complementary methods. Z-score outliers (z) were identified using a threshold of ±3, which was selected to flag extreme deviations while minimizing false positives. Isolation Forest outliers (IF) were detected using a machine learning-based anomaly detection model [11]. The Isolation Forest algorithm was applied with default parameters (n_estimators=100, contamination='auto'), supporting a balance between generalisability and sensitivity across diverse meter types.

Data quality outputs include metrics on data completeness, outlier detection, and irregularity incidence rate, presented in the following Table D. The table holds summaries of the average metrics across all meters and is also presented in radar charts as average metric per building in Figure 3 and weather station in Figure 4.

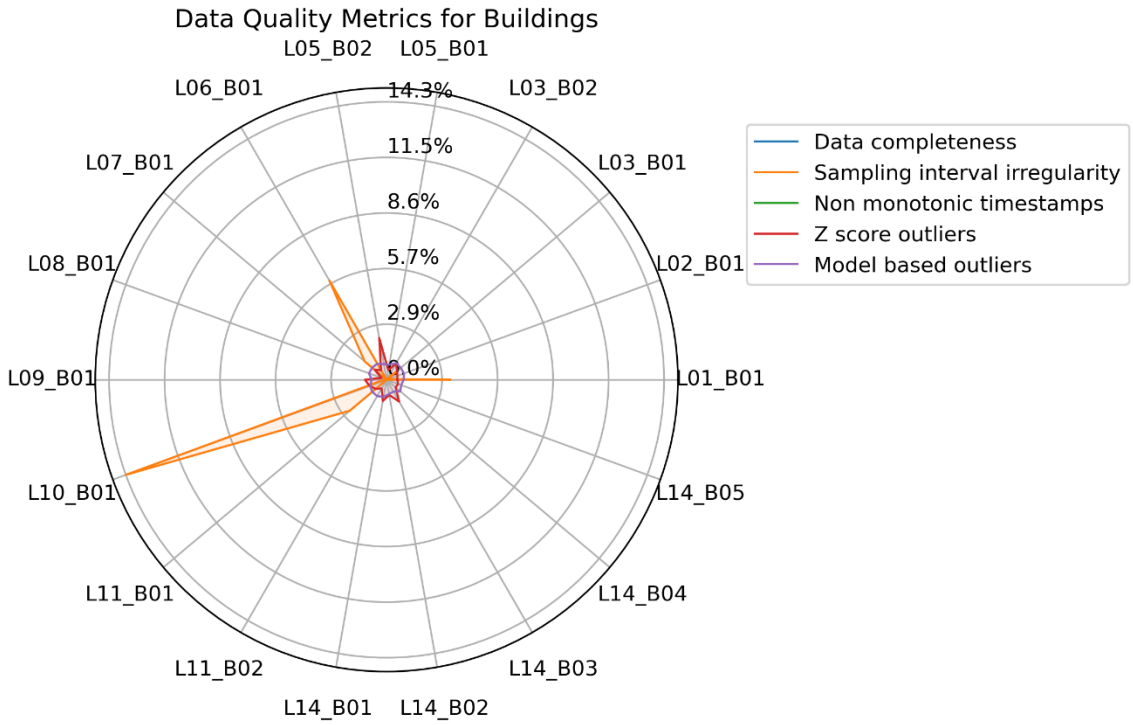


Figure 3. Radar chart of data quality metrics per building.

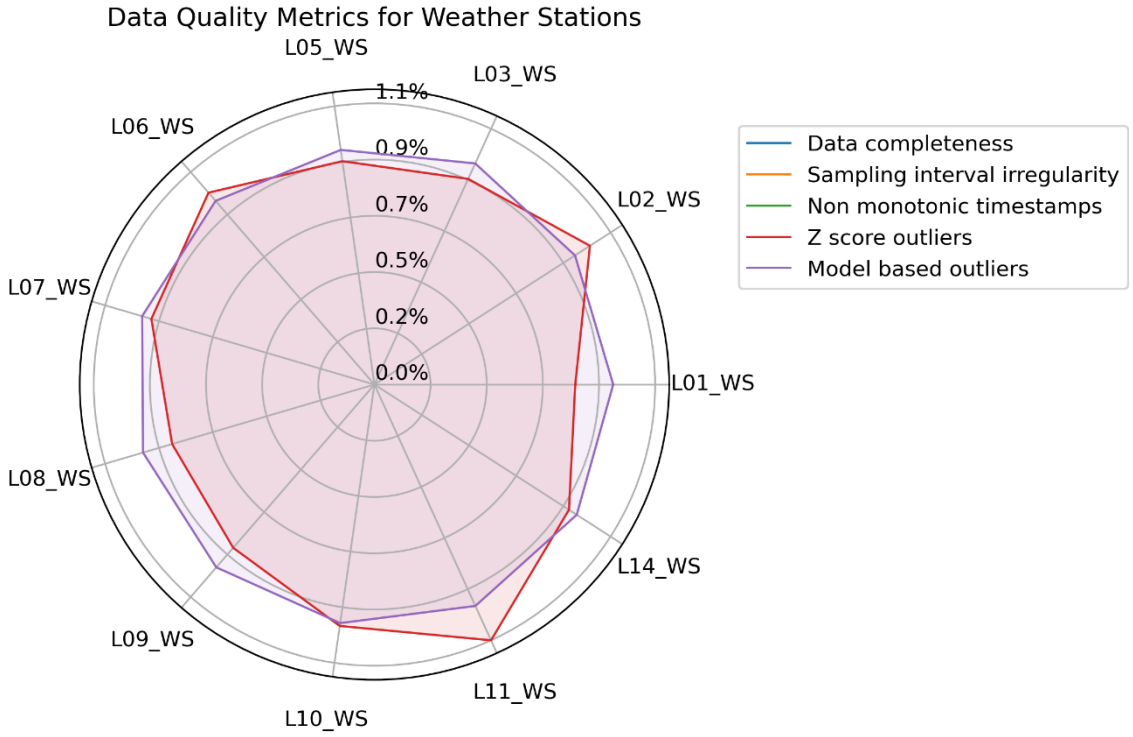


Figure 4. Radar chart of data quality metrics per weather station.

Table IV. Summary of average metrics across all meters in the ADRENALIN dataset.

Metric	***Description***	***Abbreviation***	***Avg. Rate (%)***
Missing Values	Gaps in the time series	MISS	> 0.0
Non monotonic timestamps	Preservation of timestamp order	NMT	> 0.0
Time Step Irregularity	Inconsistent time intervals	TSI	0.99
Z-score Outliers	Statistical anomalies	z	0.88
Isolation Forest Outliers	ML-based anomaly detection	IF	0.92
Total Irregularity Rate	Combined metric	—	**2.79**

The overall irregularity rate across all recorded data was approximately 2.8%, based on the detection of missing value, Time step irregularities, z-score outliers, and Isolation Forest outliers – calculated by Equation 1 below.

$$Average\ Irregularities\ Rate = \frac{\sum(MISS + NMT + TSI + z + IF\ )}{\sum Data\ Points} \qquad (1)$$

Limitations

Building energy data timeseries may exhibit gaps or missing periods, which can occur when measuring equipment experiences offline periods. Additionally, data might be recorded in incorrect units due to errors in labelling within building management systems or adjustments made to meters. Miscommunication during data collection processes, such as misunderstanding the energy system, a meter's function, or its purpose in quantifying energy streams, may also contribute to inaccuracies or inconsistencies. Due to the presence of missing or irregularly spaced data points, users should exercise caution when down-sampling the dataset, since down-sampling without addressing gaps may distort trends, especially in irregular or event-based time series, as presented in Figure 3.

Furthermore, building providers often adopt different strategies for collecting energy data, which can lead to variations in data quality and comparability when analysing multiple buildings. Another factor affecting the dataset is whether meters record data as accumulated values over time or as instantaneous measurements, as this distinction can significantly influence the interpretation and use of the data.

Additionally, some buildings' meters use a value-change-based or event-triggered recording strategy, where data are recorded only when a change occurs. This results in irregular time series that deviate from fixed-interval sampling.

CRediT AUTHOR STATEMENT

Gustaf Bengtsson: Writing - Original draft, Conceptualization, Methodology, Data curation, Visualization, Software, Validation, Supervision

Rafael Gomez Garcia: Writing – Original draft, Conceptualization, Methodology, Investigation, Data curation, Supervision

Matt Amos: Data Management, Data curation, Software, Validation

Akram Hameed: Writing – Reviewing and Editing, Supervision, Data Curation

Synne Krekling Lien: Data curation, Writing- Reviewing and Editing

Sofia Stensson: Conceptualization, Supervision, Project administration, Data curation

Clayton Miller: Writing - Review & Editing, Conceptualization

Igor Sartori: Data curation, Writing - Reviewing and Editing, Project administration

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